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TO:

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SUBJECT: Report on Meeting of 12-5-56, Automatic Programming for Sage System

December 7, 1956

Purpose:

The purpose of this meeting was to review the present programming problems of the Sage System and to discuss possible programming requirements and techniques for alleviating the problem.

Discussion:

Mr. J. E. Hamlin presented the programming problem confronting the Sage system. He pointed out that the operation has grown from a 10 man effort on a 10,000 instruction program to approximately 500 men on a 75,000 instruction program.

It was shown that the steps for programming any application could be stated as follows:

1. Recognize Problem
2. Draw Flow Charts
3. Code Program
4. Debug Program

It was then pointed out that 350 man months have been expended on step one, 250 on step two, 150 on step three and 800 on step four.

Continued discussion of the problem led to the following conclusions:

1. One of the main areas of trouble was in the inter-relationship of programs when errors were discovered.

2. A rigid set of programming notation was needed.

3. Problems were compounded by the small memory size for the problem.

4. The machine was lacking in logical ability to properly handle the application.

5. An automatic programming system would be helpful to the Sage system, however, present procedures will have to be cleaned up before a workable system could be undertaken.

Recommendations:

1. Immediately draw up a framework of a new rigid notation system.

2. Write a checking program to determine what other programs are affected by a change in any given program.

3. Expand the compiler and utilize macro instructions as an effort towards more efficient programming.

4. Add additional logic and possibly macro instructions to the present machine.

5. Several suggested programming techniques which could be employed to make more efficient use of memory were generative routines and on-the-spot compiling.

6. Arrange to have the MIT people come here to give more detailed information on the problem and discuss the recommendations for solution. This will be arranged by Mr. Hamlin.

D. F. Milvaney

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cc: Dr. R. J. Cypser
Mr. N. P. Edwards